Is rbsD (b3748) required for growth of *E. coli* iML1515 on D-ribose?

1. Summary (answer)

The two model variants disagree. In the **published iML1515**, a `b3748` (rbsD) single-gene deletion has **no effect** on predicted growth on D-ribose (KO/WT = 1.00) — rbsD is dispensable. In the **reannotated variant**, where rbsD is represented by its UniProt-annotated function (D-ribose pyranase, EC 5.4.99.62) instead of as a ribose-transporter subunit, the deletion is **lethal** on ribose (KO/WT = 0.00) — rbsD is strictly essential. The reannotated result is the one consistent with rbsD's known biochemistry.

2. Result table

Variant	WT growth (h^{-1})	b3748-KO growth (h^{-1})	KO/WT
Glucose minimal baseline (published WT)	**0.876997**	–	–
(a) Published iML1515, ribose minimal	**0.688913**	**0.688913**	**1.000**
(b) Reannotated (rbsD = pyranase), ribose minimal	**0.688913**	**0.000000**	**0.000**

No-carbon control (all C-source uptake closed): growth = 0.000000 h^{-1} — confirms no hidden carbon source and that the ribose result is genuinely ribose-dependent.

3. What was inspected / modified

Where b3748 lives in the published model. `b3748` appears in exactly **one** GPR: `RIBabcpp` (D-ribose ABC transport, periplasm→cytoplasm). Its GPR is a logical OR of three alternative transporter complexes:

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(b4230 and b4227 and b4485 and b4231) or (b3750 and b3751 and b3749 and b3748) or (b4088 and b40
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Because b3748 sits inside just one of three OR branches, deleting it alone leaves `RIBabcpp` fully functional (0 reactions killed) → the published KO grows normally. Ribokinase `RBK` (`rib__D_c → r5p_c`, GPR `b3752`) is untouched by the deletion.

Reannotation edit (variant b), one defensible implementation: 1. Added metabolite `rib__D_furan_c` (β-D-ribofuranose, $C5H10O5$). 2. Added reaction `RBSD_pyranase`: `rib__D_c ⇌ rib__D_furan_c`, bounds [−1000, 1000], **GPR** = **`b3748`** (reversible isomerisation, EC 5.4.99.62). 3. Rerouted `RBK` to consume the pyranase product: before `{atp_c:-1, rib__D_c:-1, adp_c:1, h_c:1, r5p_c:1}`; after `{atp_c:-1, rib__D_furan_c:-1, adp_c:1, h_c:1, r5p_c:1}`. 4. Removed `b3748` from the `RIBabcpp` GPR (rbsD is not a transporter subunit): the affected branch `(b3750 and b3751 and b3749 and b3748)` → `(b3750 and b3751 and b3749)`.

After the edit, `b3748` appears **only** in `RBSD_pyranase`, which becomes the sole route from cytosolic ribose to the ribokinase substrate. Deleting b3748 kills exactly `['RBSD_pyranase']`, severing ribose → R5P → biomass, so growth collapses to 0.

Medium (explicit). Started from the inorganic exchanges that are open by default in the distributed model (Ca, Cl, CO_2, Co, Cu, $Fe^{2+}/^{3+}$, H_2O, H^+, K, Mg, Mn, MoO_4, Na, NH_4, Ni, O_2, Pi, selenate, selenite, SO_4, tungstate, Zn — all lb = −1000; O_2 open = aerobic). For **glucose** medium: `EX_glc__D_e` lb = −10. For **ribose** medium: `EX_glc__D_e` lb = 0 and `EX_rib__D_e` lb = −10. The only carbon-containing exchange left open in the ribose medium is `EX_rib__D_e` (plus inert `EX_co2_e`, which cannot support heterotrophic growth — verified by the zero no-carbon control). Gene deletion was done via GPR boolean evaluation (b3748 = False → knock out any reaction whose GPR flips True→False), not ad-hoc bound edits; the original model object was never mutated (variant b is a deep copy).

4. Interpretation vs. experiment

The **published model's prediction (KO grows = WT) does not match the biochemistry** of rbsD. UniProt (P04982) and structural/enzymatic studies identify RbsD as the **D-ribose pyranase** that converts imported β-D-ribopyranose to the β-D-ribofuranose anomer, "the key step for substrate supply to ribokinase RbsK" (PMID **21276853**). Ribose enters the cell predominantly as the pyranose, but ribokinase acts on the furanose; without the pyranase the cell depends only on slow spontaneous mutarotation. Comparative genomics reinforces the catabolic role: lactic-acid-bacterial genomes lacking rbsD show **"no growth … on … ribose, as sole carbon source … referred to the absence of ribose pyranase rbsD"** (PMID **33129664**).

Critically, the primary *E. coli* literature makes exactly this functional split: in the *rbsDACBK* operon, **"RbsABC forms the ABC-type high-affinity d-ribose transporter, while RbsD and RbsK are involved in the conversion of d-ribose into d-ribose 5-phosphate"** (Shimada, Kori & Ishihama 2013, PMID **23651393**). That is, RbsD is a *catabolic enzyme*, not a transporter subunit — precisely the assignment the published iML1515 gets wrong and the reannotation restores.

So the **reannotated variant is the one that matches the annotated/experimental phenotype**: rbsD contributes to ribose catabolism (ribose → R5P), whereas the published iML1515 mis-files it as a redundant transporter subunit and therefore wrongly calls it dispensable. Caveat: the reannotated model predicts *absolute* lethality (KO = 0), while spontaneous anomerisation means a real Δ*rbsD* strain likely retains *slow* ribose growth rather than none — i.e. the true phenotype is a strong growth defect, which the base FBA idealises to zero (quantified below).

How the result changes under alternative edits (all numbers are real COBRApy/GLPK solver calls; see `cobra_alternatives.py` / `cobra_alternatives_output.txt`).

Implementation	WT	KO	KO/ WT
Chosen: RBK anomer-specific, no spontaneous route	0.688913	0.000000	0.000
ALT 1: pyranase added but RBK still accepts `rib__D_c` (pyranase optional)	0.688913	0.688913	1.000
ALT 2: RBK anomer-specific + gene-less spontaneous mutarotation, cap = 0.5	0.688913	0.009388	0.014
ALT 2: cap = 1.0	0.688913	0.045261	0.066
ALT 2: cap = 2.0	0.688913	0.117006	0.170
ALT 2: cap = 5.0	0.688913	0.331554	0.481
ALT 2: cap = ∞ (unbounded)	0.688913	0.688913	1.000

The essentiality result **hinges on ribokinase being anomer-specific**: if `RBK` is left able to phosphorylate `rib__D_c` directly (ALT 1), the pyranase carries no obligatory flux and the KO is again non-lethal (KO/WT = 1). I chose the obligatory-coupling implementation because it reflects the biochemical rationale (RbsK acts on the furanose that only RbsD supplies) that motivates the reannotation. ALT 2 shows that adding a slow non-enzymatic mutarotation converts the absolute lethality into a graded growth defect — the biologically realistic picture of a Δ*rbsD* strain that grows slowly on ribose rather than not at all.

5. Provenance

- **Model:** iML1515 v1, `http://bigg.ucsd.edu/static/models/iML1515.json`, retrieved 2026-08-12. Size 3,062,491 bytes. md5 `9579313bc1458acf4ef0ec44bf852ede`, sha256 `b0f9199f048779bb08a14dfa6c09ec56d35b8750d2f99681980d0f098355fbf5` (computed via `md5sum`/`sha256sum`). 2712 reactions, 1877 metabolites, 1516 genes.
- **COBRApy (primary, as requested): cobra 0.32.1**, solver **GLPK via optlang (`optlang.glpk_interface`)**. COBRApy is not importable inside the OpenScientist `execute_code` sandbox (import allowlist), so it was `pip install`-ed and the analysis executed in a **subprocess**. It uses COBRApy's own gene-deletion facilities — `model.genes.get_by_id("b3748").knock_out()` inside a `with model:` context **and** `cobra.flux_analysis.single_gene_deletion` — and never mutates the loaded model across experiments (context managers + `model.copy()` for the reannotated variant). Artifacts: `cobra_iML1515_rbsD.py`, `cobra_output.txt`.
- **Independent cross-check:** a hand-rolled LP (`scipy.optimize.linprog(method="highs")`, HiGHS, scipy 1.17.1) built directly from the BiGG JSON ($S \cdot v = 0$, $lb \le v \le ub$, maximise `BIOMASS_Ec_iML1515_core_75p37M`). **It agrees with COBRApy to 6 decimals** on every number below (glucose 0.876997; ribose WT 0.688913; published KO 0.688913; reannotated KO 0.000000), and independently reproduces the accepted iML1515 glucose baseline. Artifacts: `fba_rbsD_iML1515.py`, `fba_rbsD_output.txt`.
- **Sensitivity analysis artifacts:** `cobra_alternatives.py`, `cobra_alternatives_output.txt`.

Number	COBRApy 0.32.1 (GLPK)	scipy linprog (HiGHS)
Glucose baseline WT	0.876997	0.876997
Ribose WT (both variants)	0.688913	0.688913
Published ribose KO	0.688913	0.688913
Reannotated ribose KO	0.000000 (infeasible)	0.000000 (infeasible)

6. Supported / refuted hypotheses

- **Supported:** rbsD is dispensable in published iML1515 (KO/WT = 1.0); rbsD is essential when modelled as ribose pyranase (KO/WT = 0.0); the two variants disagree.
- **Refuted:** that the published iML1515 correctly captures rbsD's metabolic role on ribose — it does not (it treats rbsD as a redundant transporter subunit).

7. Limitations & future directions

- FBA gives a binary growth/no-growth idealisation; it cannot capture the partial (kinetically slowed) phenotype expected from residual spontaneous mutarotation.
- The functional assignment (RbsD = catabolic ribose→R5P enzyme, not a transporter subunit) is directly supported by primary *E. coli* work (PMID **23651393**), the pyranase mechanism (PMID **21276853**), and a comparative-genomics growth association (PMID **33129664**). A dedicated growth-curve of a clean Δ*rbsD* strain on ribose (e.g. Ryu/Kim, *J. Biol. Chem.* 2004, not retrievable through this PubMed index) would further pin down the magnitude of the residual (mutarotation-dependent) growth that ALT 2 above brackets.
- Extending the reannotation with an explicit non-enzymatic mutarotation reaction (small vmax) would let the model reproduce a *slow* rather than zero Δ*rbsD* growth.